

REI603M - User Testing Guide

A Companion Guide for Testing AI-Powered Applications

This guide is not graded. It helps you with the user testing section of your final project. Read it before you start testing.

1. Why Test with Real Users

Your Assignment 3 evaluation dataset tests whether the AI produces correct outputs. User testing answers a different question: can a person actually use your product?

Automated evaluation will not tell you that users misread your interface, phrase requests in ways your prompts handle poorly, or give up after three seconds of waiting for a response. Five users sitting in front of your application will.

In A3, you planned what interaction data to collect. In A6, you built the collection infrastructure. Now you close the loop by putting real people in front of the product.

If you only test with your group members, you will miss every assumption you share.

2. Planning Your Test

2.1. Define What You Are Testing

Pick 2–3 specific questions you want answered. Not “does it work” but “can a first-time user complete the primary flow without help” or “do users trust the AI output enough to act on it.”

Research Question	What It Reveals
Can a user complete the primary flow without help?	Usability gaps, confusing UI elements
Do users understand what the AI output means?	Output format or explanation problems
What do users try that you did not expect?	Missing features, wrong assumptions
Does the response time feel acceptable?	Latency tolerance for your use case
Do users trust the output?	Calibration and explanation needs

2.2. Design Tasks, Not Scripts

Give users goals, not step-by-step instructions. “Find a recipe for a vegetarian dinner for 4 people” works. “Type ‘vegetarian dinner’ into the search box” does not. The first version lets you see how users approach your product. The second tests whether they can follow orders.

Write 3–5 realistic tasks that cover your primary user flow. Each task should take 2–5 minutes. Include at least one task that will expose a known limitation. You want to see how users react when the AI fails.

2.3. Prepare Your Data Collection

Go back to your A3 interaction data plan and your A6 analytics setup. Is the logging infrastructure working? At minimum, capture each test session's inputs, outputs, timestamps, and errors.

Decide how you will collect qualitative data. Your options: think-aloud protocol (user narrates while using), observation notes, post-session interview questions, a short questionnaire, or a combination.

The simplest approach that works: sit next to the user, take notes, and ask 3 questions afterward. Do not over-engineer this.

2.4. Pre-Testing Checklist

Before your first session, verify:

- Application is deployed and accessible (not running on localhost)
- Observability and logging are active (Langfuse, PostHog, or your custom setup)
- You have 3–5 written task descriptions
- You know your 2–3 research questions
- You have a way to record notes (document, spreadsheet, or paper)
- You have run through all tasks yourself to confirm they work
- API keys have sufficient credits for all test sessions
- You have a brief intro script ready (see next section)

3. Running Sessions

3.1. Recruiting 5+ Users

Anyone who is not in your group counts. Classmates from other groups, friends, family, strangers. The closer they are to your actual target users, the more useful the feedback. But any external perspective beats none.

A practical option: pair with another group in the class and test each other's applications. Both groups benefit.

Five users typically surface around 80% of major usability problems. More is better, but five is sufficient for the final project.

3.2. Session Structure

Keep sessions short. Twenty minutes is enough.

Phase	Duration	What You Do
Introduction	1–2 min	Explain that you are testing the product, not the user. There are no wrong answers. Ask permission to take notes.

Tasks	10–15 min	Give tasks one at a time. Observe. Take notes. Resist the urge to help.
Debrief	3–5 min	Ask open-ended questions about what was confusing, surprising, or frustrating.

3.3. What to Observe

Watch for:

- Where the user hesitates or pauses
- What they say out loud: confusion, surprise, frustration
- Whether they understand the AI output or need to re-read and retry
- Unexpected inputs or workflows they attempt
- Points where the AI output was wrong, misleading, or unhelpful
- Whether the user trusts the output or double-checks it elsewhere

3.4. AI-Specific Observations

Testing AI applications differs from testing traditional software. Pay attention to:

Non-deterministic output. The same input may produce different results across sessions. Note when this happens and whether it confuses the user.

Latency. LLM calls take seconds. Watch whether users think the app is broken during generation. Do they wait patiently, click again, or leave?

Prompt sensitivity. Users will phrase things differently than your test cases. Note inputs that produce poor results and compare them to your A3 evaluation dataset.

Overreliance. Some users trust AI output without question. Note when users should be skeptical but are not.

Hallucination reactions. When the AI produces confident but wrong output, does the user notice? This is critical data for your error analysis.

Recovery from failure. When the AI gives a bad response, can the user recover? Do they retry, rephrase, or give up?

Do not help the user. When they struggle, take notes. If they are completely stuck for more than 60 seconds, give a small hint and record that you intervened.

4. Analyzing Results

4.1. Organize Your Notes

After each session, spend 5 minutes writing your top 3 observations while they are fresh. Do not wait until all sessions are done.

Create a simple table: one row per finding, with columns for which users experienced it, severity (blocks task / causes confusion / minor friction), and category (UI, AI output quality, latency, missing feature, trust).

4.2. Look for Patterns

A problem one user hit might be an outlier. A problem three users hit is real. Rank findings by frequency and severity.

The most useful findings are the ones you did not anticipate. If you already knew about a problem, user testing confirmed it. If something surprises you, that is where you learn.

4.3. Connect Interaction Data to Observations

Cross-reference your notes with the logged data from your analytics setup. Look at Langfuse traces (or equivalent) from the test sessions. The traces may reveal things the user did not mention: multiple retries, high latency on specific queries, token usage spikes, or error codes the user never saw.

The final project grading asks you to “show interaction data you collected and what it revealed about usage patterns.” This is where you build that evidence.

4.4. Findings Template

Use this structure to organize each finding for your report:

Finding: One-sentence description
Users affected: e.g., 4 out of 5
Severity: Blocks task / Causes confusion / Minor friction
Category: UI / AI output / Latency / Missing feature / Trust
Evidence: What you observed + supporting interaction data
Action taken: What you changed or plan to change

5. Turning Findings into Improvements

5.1. Prioritize

You will not fix everything before the final submission. Pick the 3–5 most impactful findings. Fix blockers first (things that prevent task completion), then confusion, then friction.

Some fixes are prompt changes that take minutes. Others are UI changes that take hours. Factor effort into your prioritization.

5.2. Close the Loop

Your final report needs to show how user feedback informed improvements. For each finding you acted on: describe what you observed, what you changed, and if possible, whether the change helped.

You do not need to re-test with all 5 users after making changes. Even re-testing with 1–2 users strengthens the report.

5.3. What to Include in Your Report

Your final project report (Section 7) must cover:

- **Testing methodology:** what you did, how you recruited, what tasks you gave
- **Findings:** what worked, what confused users, what broke
- **Interaction data:** what your logging and analytics revealed

- **Improvements:** how feedback changed the product

6. Quick Reference

6.1. Session Facilitation

Do	Don't
Say "What are you thinking right now?"	Say "You should click that button"
Let silence happen. Users are processing.	Fill every pause with explanations
Ask "What did you expect to happen?"	Ask "Did you like it?" (leading question)
Note exact quotes from users	Paraphrase or interpret on the fly
Test the full flow beforehand	Discover bugs during the session
Thank the user afterward	Make the user feel tested or judged

6.2. Post-Session Questions

Five questions for the debrief:

1. What was the most confusing part?
2. Was there a point where you were not sure what to do next?
3. How did you feel about the AI's responses? Were they what you expected?
4. Would you use this for *the real task it is designed for*? Why or why not?
5. What would you change?

6.3. Resources

- Steve Krug, *Rocket Surgery Made Easy* – the standard reference for lightweight user testing. The approach in this guide draws from Krug's methodology.
- [Nielsen Norman Group: Usability Testing 101](#) – comprehensive overview of usability study methods
- [Jakob Nielsen: Why You Only Need to Test with 5 Users](#) – the research behind the 5-user recommendation